

Data Base Management System (DBMS)

09/09/2026

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Roll No : 15

Batch : B1

Assignment No 4 – SQL Agregate Functions

Question 1: Employee Management

Table Structure:

EmployeeDetails(EmpID, EmpName, MonthlySalary, City)

Perform the following SQL queries

Queries:

- a) Display all employees who belong to the **IT department**.

```
mysql> select * from EmployeeDetails where Department='IT';
+-----+-----+-----+-----+-----+
| EmpID | EmpName      | Department | MonthlySalary | City |
+-----+-----+-----+-----+-----+
|      1 | Harshad Teli | IT         |      60000    | KOP  |
|      2 | Anurag Lad   | IT         |      45000    | Pune |
+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

- b) Display employees whose **MonthlySalary is greater than 30000**.

```
mysql> select * from EmployeeDetails where MonthlySalary > 30000;
+-----+-----+-----+-----+-----+
| EmpID | EmpName      | Department | MonthlySalary | City   |
+-----+-----+-----+-----+-----+
|      1 | Harshad Teli | IT         |      60000    | KOP    |
|      2 | Anurag Lad   | IT         |      45000    | Pune   |
|      3 | Vardhan Patki | Marketing  |      35000    | Satara |
|      4 | Kartika Phalle | IT         |      75000    | Karad  |
+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql>
```

c) Display employees who belong to **HR or Finance** department.

```
mysql> select * from EmployeeDetails where Department='HR' OR Department='Finance';
+-----+-----+-----+-----+-----+
| EmpID | EmpName      | Department | MonthlySalary | City   |
+-----+-----+-----+-----+-----+
|      4 | Kartika Phalle | Finance    |      75000    | Karad  |
|      5 | Anjali Patil   | HR         |      25000    | Mumbai |
+-----+-----+-----+-----+-----+
2 rows in set (0.00 sec)
```

d) Display employees whose MonthlySalary is between **25000 and 50000**.

```
mysql> select * from EmployeeDetails where MonthlySalary BETWEEN 25000 AND 50000;
+-----+-----+-----+-----+-----+
| EmpID | EmpName      | Department | MonthlySalary | City   |
+-----+-----+-----+-----+-----+
|      2 | Anurag Lad    | IT         |      45000    | Pune   |
|      3 | Vardhan Patki | Marketing  |      35000    | Satara |
|      5 | Anjali Patil   | HR         |      25000    | Mumbai |
+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)
```

e) Display all employee records in **ascending order of EmpName**.

```
mysql> select * from EmployeeDetails order by EmpName;
+-----+-----+-----+-----+-----+
| EmpID | EmpName      | Department | MonthlySalary | City   |
+-----+-----+-----+-----+-----+
|      5 | Anjali Patil   | HR         |      25000    | Mumbai |
|      2 | Anurag Lad     | IT         |      45000    | Pune   |
|      1 | Harshad Teli   | IT         |      60000    | KOP    |
|      4 | Kartika Phalle | Finance    |      75000    | Karad  |
|      3 | Vardhan Patki  | Marketing  |      35000    | Satara |
+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

f) Display all employee records in **descending order of MonthlySalary**.

```
mysql> select * from EmployeeDetails order by MonthlySalary DESC;
+-----+-----+-----+-----+-----+
| EmpID | EmpName      | Department | MonthlySalary | City   |
+-----+-----+-----+-----+-----+
|      4 | Kartika Phalle | Finance    |      75000    | Karad  |
|      1 | Harshad Teli   | IT         |      60000    | KOP    |
|      2 | Anurag Lad     | IT         |      45000    | Pune   |
|      3 | Vardhan Patki  | Marketing  |      35000    | Satara |
|      5 | Anjali Patil   | HR         |      25000    | Mumbai |
+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

g) Display the **total number of employees** in the table.

```
mysql> select * from EmployeeDetails;
+-----+-----+-----+-----+
| EmpID | EmpName      | Department | MonthlySalary | City   |
+-----+-----+-----+-----+
|      1 | Harshad Teli | IT         |      60000    | KOP   |
|      2 | Anurag Lad   | IT         |      45000    | Pune  |
|      3 | Vardhan Patki | Marketing  |      35000    | Satara |
|      4 | Kartika Phalle | Finance    |      75000    | Karad  |
|      5 | Anjali Patil  | HR         |      25000    | Mumbai |
+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

```
mysql> select count(*) as Total_Employee from EmployeeDetails;
+-----+
| Total_Employee |
+-----+
|              5 |
+-----+
1 row in set (0.00 sec)
```

h) Find the **total monthly salary** paid to all employees.

```
mysql> select SUM(MonthlySalary) as Total_Salary from EmployeeDetails;
+-----+
| Total_Salary |
+-----+
|      240000 |
+-----+
1 row in set (0.00 sec)

mysql>
```

i) Find the **average monthly salary** of all employees.

```
mysql> select AVG(MonthlySalary) as Average_Salary from EmployeeDetails
+-----+
| Average_Salary |
+-----+
|    48000.0000 |
+-----+
1 row in set (0.03 sec)

mysql>
```

j) Find the **highest monthly salary** among all employees.

```
mysql> select MAX(MonthlySalary) as Maximum_Salary from EmployeeDetails;
+-----+
| Maximum_Salary |
+-----+
|          75000 |
+-----+
1 row in set (0.00 sec)
```

k) Find the **lowest monthly salary** among all employees.

```
mysql> select MIN(MonthlySalary) as Maximum_Salary from EmployeeDetails;
+-----+
| Maximum_Salary |
+-----+
|          25000 |
+-----+
1 row in set (0.00 sec)
```

L) Find the **number of employees in each department**.

```
mysql> select * from EmployeeDetails;
+-----+-----+-----+-----+-----+
| EmpID | EmpName      | Department | MonthlySalary | City   |
+-----+-----+-----+-----+-----+
| 1     | Harshad Teli | IT         | 60000         | KOP    |
| 2     | Anurag Lad   | IT         | 45000         | Pune   |
| 3     | Vardhan Patki | Marketing  | 35000         | Satara |
| 4     | Kartika Phalle | Finance    | 75000         | Karad  |
| 5     | Anjali Patil | HR         | 25000         | Mumbai |
+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> select count(EmpID) as NumberofEmployees,Department  from EmployeeDetails group by Department;
+-----+-----+
| NumberofEmployees | Department |
+-----+-----+
| 2                 | IT         |
| 1                 | Marketing  |
| 1                 | Finance    |
| 1                 | HR         |
+-----+-----+
4 rows in set (0.00 sec)
```

M) Find the **average monthly salary** in each department.

```
mysql> select AVG(MonthlySalary) as AverageSalary,Department from EmployeeDetails group by Department;
+-----+-----+
| AverageSalary | Department |
+-----+-----+
| 52500.0000    | IT         |
| 35000.0000    | Marketing  |
| 75000.0000    | Finance    |
| 25000.0000    | HR         |
+-----+-----+
4 rows in set (0.01 sec)
```

N) Find the **total monthly salary** in each department.

```
mysql> select MAX(MonthlySalary) as TotalSalary,Department from EmployeeDetails group by Department;
+-----+-----+
| TotalSalary | Department |
+-----+-----+
| 60000      | IT         |
| 35000      | Marketing  |
| 75000      | Finance    |
| 25000      | HR         |
+-----+-----+
4 rows in set (0.00 sec)
```

o) Display only those departments having more than 2 employees.

```
mysql> select * from EmployeeDetails;
+-----+-----+-----+-----+-----+
| EmpID | EmpName      | Department | MonthlySalary | City      |
+-----+-----+-----+-----+-----+
| 1     | Harshad Teli | IT         | 60000         | KOP       |
| 2     | Anurag Lad   | IT         | 45000         | Pune      |
| 3     | Vardhan Patki | Marketing  | 35000         | Satara    |
| 4     | Kartika Phalle | Finance    | 75000         | Karad     |
| 5     | Anjali Patil | HR         | 25000         | Mumbai    |
| 6     | Anjali Patil | IT         | 85000         | ISHWARPUR |
+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)

mysql> select Department,count(EmpID) as NumberofEmployee from EmployeeDetails GROUP BY Department HAVING count(*)>2;
+-----+-----+
| Department | NumberofEmployee |
+-----+-----+
| IT         | 3                |
+-----+-----+
1 row in set (0.00 sec)

mysql> █
```

p) Display departments where the average monthly salary is greater than 30000.

```
mysql> select Department,AVG(MonthlySalary) as AverageSalary from EmployeeDetails GROUP BY Department HAVING AVG(MonthlySalary)>30000;
+-----+-----+
| Department | AverageSalary |
+-----+-----+
| IT         | 63333.3333    |
| Marketing  | 35000.0000    |
| Finance    | 75000.0000    |
+-----+-----+
3 rows in set (0.00 sec)

mysql> █
```

Q) Display all the different cities available in the EmployeeDetails table.

```
mysql> select DISTINCT City from EmployeeDetails;
+-----+
| City   |
+-----+
| KOP    |
| Pune   |
| Satara |
| Karad  |
| Mumbai |
| ISHWARPUR |
+-----+
6 rows in set (0.00 sec)
```

R) Find the number of employees working in the IT department.

```
mysql> select * from EmployeeDetails;
+-----+-----+-----+-----+-----+
| EmpID | EmpName      | Department | MonthlySalary | City       |
+-----+-----+-----+-----+-----+
| 1     | Harshad Teli | IT         | 60000         | KOP        |
| 2     | Anurag Lad   | IT         | 45000         | Pune       |
| 3     | Vardhan Patki | Marketing  | 35000         | Satara     |
| 4     | Kartika Phalle | Finance    | 75000         | Karad      |
| 5     | Anjali Patil | HR         | 25000         | Mumbai     |
| 6     | Anjali Patil | IT         | 85000         | ISHWARPUR  |
+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)

mysql> select count(EmpID) as NumberofEmployee from EmployeeDetails Where Department='IT';
+-----+
| NumberofEmployee |
+-----+
| 3                |
+-----+
1 row in set (0.00 sec)

mysql>
```

S) Find the highest monthly salary among IT employees.

```
mysql> select MAX(MonthlySalary) as HighestSalary from EmployeeDetails Where Department='IT';
+-----+
| HighestSalary |
+-----+
| 85000         |
+-----+
1 row in set (0.00 sec)

mysql>
```

t) Display the **department-wise employee count and average salary**, showing only departments having more than 1 employee, arranged by employee count in descending order.

```
mysql> SELECT Department, COUNT(*) AS NumberofEmployee, AVG(MonthlySalary) AS Average_Salary FROM EmployeeDetails GROUP BY Department HAVING COUNT(*) > 1 ORDER BY NumberofEmployee DESC;
```

Department	NumberofEmployee	Average_Salary
IT	3	63333.3333

```
1 row in set (0.00 sec)
```

```
mysql>
```

Question 2: Student Management

Table Structure:

StudentMaster(StudentID, StudentName, Course, Age, Email)

Perform the following SQL operations:

Queries:

1) Display all students who are studying **MCA**.

```
mysql> select * from StudentMaster Where Course='MCA';
+-----+-----+-----+-----+-----+
| StudentID | StudentName | Course | StudentAge | Email |
+-----+-----+-----+-----+-----+
| 1002 | Shree Patil | MCA | 23 | shree@gmail.com |
+-----+-----+-----+-----+-----+
1 row in set (0.00 sec)

mysql>
```

2) Display students whose **StudentAge** is greater than 20.

```
mysql> select * from StudentMaster Where StudentAge>20;
+-----+-----+-----+-----+-----+
| StudentID | StudentName | Course | StudentAge | Email |
+-----+-----+-----+-----+-----+
| 1002 | Shree Patil | MCA | 23 | shree@gmail.com |
| 1003 | Shreya Vibhute | MBA | 21 | shreya@gmail.com |
| 1004 | Neha Sutar | BSC | 26 | neha@gmail.com |
| 1005 | Anurag Lad | BCA | 21 | anurag@gmail.com |
+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql>
```

3) Display students studying **MCA or BCA**.

```
mysql> select * from StudentMaster Where Course='MCA' OR Course='BCA';
+-----+-----+-----+-----+-----+
| StudentID | StudentName | Course | StudentAge | Email |
+-----+-----+-----+-----+-----+
| 1001 | Anadhut Phalle | BCA | 20 | avadhut@gmail.com |
| 1002 | Shree Patil | MCA | 23 | shree@gmail.com |
| 1005 | Anurag Lad | BCA | 21 | anurag@gmail.com |
+-----+-----+-----+-----+-----+
3 rows in set (0.00 sec)

mysql>
```


4) Display students whose age is between **18 and 25**.

```
mysql> select * from StudentMaster Where StudentAge BETWEEN 18 AND 25;
+-----+-----+-----+-----+-----+
| StudentID | StudentName | Course | StudentAge | Email |
+-----+-----+-----+-----+-----+
| 1001 | Anadhut Phalle | BCA | 20 | avadhut@gmail.com |
| 1002 | Shree Patil | MCA | 23 | shree@gmail.com |
| 1003 | Shreya Vibhute | MBA | 21 | shreya@gmail.com |
| 1005 | Anurag Lad | BCA | 21 | anurag@gmail.com |
+-----+-----+-----+-----+-----+
4 rows in set (0.00 sec)

mysql> 
```

5) Display all student records in ascending order of StudentName..

```
mysql> select * from StudentMaster order by StudentName;
+-----+-----+-----+-----+-----+
| StudentID | StudentName | Course | StudentAge | Email |
+-----+-----+-----+-----+-----+
| 1001 | Anadhut Phalle | BCA | 20 | avadhut@gmail.com |
| 1005 | Anurag Lad | BCA | 21 | anurag@gmail.com |
| 1004 | Neha Sutar | BSC | 26 | neha@gmail.com |
| 1002 | Shree Patil | MCA | 23 | shree@gmail.com |
| 1003 | Shreya Vibhute | MBA | 21 | shreya@gmail.com |
+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

6) Display all student records in **descending order of StudentAge**.

```
mysql> select * from StudentMaster order by StudentAge DESC;
+-----+-----+-----+-----+-----+
| StudentID | StudentName | Course | StudentAge | Email |
+-----+-----+-----+-----+-----+
| 1004 | Neha Sutar | BSC | 26 | neha@gmail.com |
| 1002 | Shree Patil | MCA | 23 | shree@gmail.com |
| 1003 | Shreya Vibhute | MBA | 21 | shreya@gmail.com |
| 1005 | Anurag Lad | BCA | 21 | anurag@gmail.com |
| 1001 | Anadhut Phalle | BCA | 20 | avadhut@gmail.com |
+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

7) Find the **total number of students** in the table.

```
mysql> select * from StudentMaster;
+-----+-----+-----+-----+-----+
| StudentID | StudentName      | Course | StudentAge | Email                |
+-----+-----+-----+-----+-----+
|      1001 | Anadhut Phalle   | BCA    |      20    | avadhut@gmail.com    |
|      1002 | Shree Patil      | MCA    |      23    | shree@gmail.com      |
|      1003 | Shreya Vibhute   | MBA    |      21    | shreya@gmail.com     |
|      1004 | Neha Sutar       | BSC    |      26    | neha@gmail.com       |
|      1005 | Anurag Lad       | BCA    |      21    | anurag@gmail.com     |
+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> select count(StudentID) As Total_Students from StudentMaster;
+-----+
| Total_Students |
+-----+
|              5 |
+-----+
1 row in set (0.00 sec)

mysql>
```

8) Find the **average age** of all students.

```
mysql> select AVG(StudentAge) As Average_Age from StudentMaster;
+-----+
| Average_Age |
+-----+
|    22.2000 |
+-----+
1 row in set (0.00 sec)

mysql> █
```

9) Find the **highest age** among all students.

```
mysql> select MAX(StudentAge) As Highest_Age from StudentMaster;
+-----+
| Highest_Age |
+-----+
|           26 |
+-----+
1 row in set (0.00 sec)

mysql>
```

10) Find the **lowest age** among all students.

```
mysql> select MIN(StudentAge) As Lowest_Age from StudentMaster;
+-----+
| Lowest_Age |
+-----+
|          20 |
+-----+
1 row in set (0.00 sec)

mysql>
```

11) Find the **number of students in each course**.

```
mysql> select * from StudentMaster;
+-----+-----+-----+-----+-----+
| StudentID | StudentName | Course | StudentAge | Email |
+-----+-----+-----+-----+-----+
| 1001 | Anadhut Phalle | BCA | 20 | avadhut@gmail.com |
| 1002 | Shree Patil | MCA | 23 | shree@gmail.com |
| 1003 | Shreya Vibhute | MBA | 21 | shreya@gmail.com |
| 1004 | Neha Sutar | BSC | 26 | neha@gmail.com |
| 1005 | Anurag Lad | BCA | 21 | anurag@gmail.com |
+-----+-----+-----+-----+-----+
5 rows in set (0.00 sec)

mysql> select Count(StudentID) as Total_Students,Course from StudentMaster GROUP BY Course;
+-----+-----+
| Total_Students | Course |
+-----+-----+
| 2 | BCA |
| 1 | MCA |
| 1 | MBA |
| 1 | BSC |
+-----+-----+
4 rows in set (0.00 sec)

mysql>
```

12) Find the **average age of students in each course**.

```
mysql> select AVG(StudentAge) as Average_Age,Course from StudentMaster GROUP BY Course;
+-----+-----+
| Average_Age | Course |
+-----+-----+
| 20.5000 | BCA |
| 23.0000 | MCA |
| 21.0000 | MBA |
| 26.0000 | BSC |
+-----+-----+
4 rows in set (0.00 sec)

mysql>
```

13) Find the **total number of students in each course** and arrange the result by student count in descending order.

```
mysql> select Count(StudentID) as Total_Students,Course from StudentMaster GROUP BY Course ORDER BY Count(StudentID) DESC;
+-----+-----+
| Total_Students | Course |
+-----+-----+
| 2 | BCA |
| 1 | MCA |
| 1 | MBA |
| 1 | BSC |
+-----+-----+
4 rows in set (0.00 sec)

mysql>
```

14) Display only those courses having more than 2 students.

```
mysql> select * from StudentMaster;
+-----+-----+-----+-----+-----+
| StudentID | StudentName | Course | StudentAge | Email |
+-----+-----+-----+-----+-----+
| 1001 | Anadhut Phalle | BCA | 20 | avadhut@gmail.com |
| 1002 | Shree Patil | MCA | 23 | shree@gmail.com |
| 1003 | Shreya Vibhute | MBA | 21 | shreya@gmail.com |
| 1004 | Neha Sutar | BSC | 26 | neha@gmail.com |
| 1005 | Anurag Lad | BCA | 21 | anurag@gmail.com |
| 1006 | Avani Patil | BCA | 21 | avani@gmail.com |
+-----+-----+-----+-----+-----+
6 rows in set (0.00 sec)

mysql> select Course,Count(StudentID) as Total_Students from StudentMaster GROUP BY Course HAVING Count(StudentID) >2;
+-----+-----+
| Course | Total_Students |
+-----+-----+
| BCA | 3 |
+-----+-----+
1 row in set (0.00 sec)

mysql>
```

15) Display courses where the average age is greater than 20.

```
mysql> select Course,AVG(StudentAge) as Average_Age from StudentMaster GROUP BY Course HAVING AVG(StudentAge) >20;
+-----+-----+
| Course | Average_Age |
+-----+-----+
| BCA | 20.6667 |
| MCA | 23.0000 |
| MBA | 21.0000 |
| BSC | 26.0000 |
+-----+-----+
4 rows in set (0.00 sec)

mysql>
```

16) Display all the different courses available in the StudentMaster table.

```
mysql> select DISTINCT Course from StudentMaster;
+-----+
| Course |
+-----+
| BCA |
| MCA |
| MBA |
| BSC |
+-----+
4 rows in set (0.00 sec)

mysql>
```

17) Find the **number of students studying in the MCA course.**

```
mysql> select Count(StudentID) as Total_Students from StudentMaster where Course='MCA';
+-----+
| Total_Students |
+-----+
|          1 |
+-----+
1 row in set (0.00 sec)

mysql>
```

18) Find the **average age of MCA students.**

```
mysql> select AVG(StudentAge) as Average_Age from StudentMaster where Course='MCA';
+-----+
| Average_Age |
+-----+
|    23.0000 |
+-----+
1 row in set (0.00 sec)

mysql>
```

19) Find the **highest age among MBA students.**

```
mysql> select * from StudentMaster;
+-----+-----+-----+-----+-----+
| StudentID | StudentName | Course | StudentAge | Email |
+-----+-----+-----+-----+-----+
| 1001 | Anadhut Phalle | BCA | 20 | avadhut@gmail.com |
| 1002 | Shree Patil | MCA | 23 | shree@gmail.com |
| 1003 | Shreya Vibhute | MBA | 21 | shreya@gmail.com |
| 1004 | Neha Sutar | BSC | 26 | neha@gmail.com |
| 1005 | Anurag Lad | BCA | 21 | anurag@gmail.com |
| 1006 | Avani Patil | BCA | 21 | avani@gmail.com |
| 1007 | Raj Sawant | MBA | 25 | rajs@gmail.com |
+-----+-----+-----+-----+-----+
7 rows in set (0.00 sec)

mysql> select MAX(StudentAge) as Highest_Age from StudentMaster where Course='MBA';
+-----+
| Highest_Age |
+-----+
|          25 |
+-----+
1 row in set (0.00 sec)

mysql>
```

20) Display the **course-wise student count and average age**, showing only courses having more than 1 student, arranged by student count in descending order.

```
mysql> SELECT Course, COUNT(StudentID) AS Total_Students, AVG(StudentAge) AS Average_Age FROM StudentMaster GROUP BY Course HAVING COUNT(StudentID) > 1 ORDER BY COUNT(StudentID) DESC;
```

Course	Total_Students	Average_Age
BCA	3	20.6667
MBA	2	23.0000

```
2 rows in set (0.00 sec)
```

```
mysql>
```